INORGANIC CUME

March 14, 2014

(1). Show chemical equation(s) to make an 18-e carbonyl compound of Mo and an 18-e metallocene of Mo. (20')

(2). If the metallocene is mixed with benzene and/or CO, what new and stable compound could form? (20')

(3). Use a sketch to explain the detail of the organometallic bond(s) in the carbonyl and metallocene compounds. (20')

(4). Use a Mo atom and ligand(s) of your choice to construct a compound that can explain the "meridional" structure, and then give the point group of this compound. (20')

(5). Use Mo and ligand(s) of your choice to make an 18-e diamagnetic O_h complex and an 18-e paramagnetic O_h complex (20').